# Review of "Convex Perturbation and the Suppression of the Indivisibility Deficit"

## 1 Overall assessment

This paper investigates how the Chapman–Kolmogorov deficit of an indivisible two-time column-stochastic family changes when the family is convexly mixed with a divisible reference process. Its principal contribution is an exact algebraic decomposition of the perturbed deficit into a residual substrate term and a substrate–reference interaction term. The latter permits a non-monotone response to increasing coupling, motivating the distinction between a first-crossing threshold, at which indivisibility first falls below a prescribed detection level, and a terminal threshold, beyond which it remains suppressed. The paper derives an explicit sufficient bound for terminal suppression, establishes quadratic suppression of an integrated trajectory-space deficit as the coupling approaches unity, and extends the framework to state-dependent and nonlinear couplings. Convex mixing and robustness of non-Markovianity have substantial precedents in the literature on quantum dynamical maps and resource theories; consequently, the broad principle that mixing can alter or eliminate divisibility is not itself new. The more specific novelty lies in formulating this problem for continuous two-time column-stochastic families using the operator-norm Chapman–Kolmogorov deficit, identifying the associated interaction term, and developing the two-threshold detectability framework. These constitute a coherent and potentially useful specialist contribution, although several of the principal estimates follow from elementary expansion, norm inequalities and compactness arguments, and the physical interpretation of the mixture as an "observer coupling" requires careful qualification.

## 2 Major comments

1. The abstract is technically informative but overly dense, as it attempts to include nearly every result, extension, and interpretive qualification. It would be clearer to concentrate on the exact deficit identity, the possible non-monotonicity, the resulting distinction between the first-crossing and terminal thresholds, and the principal suppression bound. Secondary material concerning state-dependent coupling, causal operators, and speed-limit comparisons could be left to the introduction.

2. The statement that "arbitrarily strong suppression is reached at a computable sub-unit coupling" should be qualified. The result provides a sufficient coupling $g_0 < 1$ for each fixed positive tolerance $\varepsilon$, whereas $g_0$ generally approaches one as $\varepsilon \to 0$. Exact suppression is guaranteed at $g = 1$, except in special cases. The distinction between an arbitrarily small positive deficit and exact vanishing should therefore be made explicit.

3. The mathematical convex family is well defined, but its interpretation as a dynamically consistent state-conditioned interaction requires further justification. The manuscript should clarify whether the substrate/reference choice is made once per trajectory, independently over successive intervals, or only at the level of empirical two-time kernels. These constructions need not generate the same composed dynamics. The author should either derive the mixture from an explicit measurement, control, or state-feedback protocol at the trajectory level, or present it more modestly as an abstract convex perturbation model.

4. The repeated description of the convex perturbation as 'canonical" appears too strong. It is a natural and particularly tractable fixed-weight model, but other possibilities include generator perturbations, sequential interventions, kernel compositions, and time- or state-dependent mixtures. Unless a representation or uniqueness result can be supplied, 'a natural minimal model" or "the simplest fixed-weight convex model" could be more appropriate.

5. The preliminary statement that the perturbed deficit is controlled by the substrate deficit weighted by the residual mass is incomplete, because Theorem 3.1 subsequently identifies an additional substrate–reference interaction term. The introduction should mention this term immediately so that the preliminary description accurately reflects the principal identity.

6. The statement that the basic observation is elementary but its quantitative consequences are not may slightly overstate the mathematical depth of the subsequent estimates. The exact identity follows by direct expansion, the suppression bound follows by the triangle inequality, and several later stability statements use continuity and compactness. The non-monotone structure and resulting two-threshold formulation are interesting, but the claims should remain proportionate to the techniques used.

7. The statement that the threshold behaviour would induce corresponding behaviour in standard non-Markovianity functionals requires qualification. Such a quantitative transfer would generally require continuity assumptions or an explicit comparison inequality between the chosen functional and $D_{\mathrm{CK}}$. Because the manuscript later acknowledges that this transfer is not established, the introductory statement should be phrased as a possible connection rather than as a consequence.

8. Remark 2.4 correctly distinguishes two-time Chapman–Kolmogorov indivisibility from general non-Markovianity. The issue is therefore not absent from the manuscript. Nevertheless, the terminology should remain consistent with that remark throughout the abstract and introduction: the results establish suppression of CK indivisibility and do not, by themselves, establish the elimination of every possible form of higher-order temporal memory.

9. The statement that the perturbation theorem "specializes, in the discrete-time multi-instrument setting, to a statement about CPTP perturbations of process tensors" requires either a precise formulation and proof or removal. A process tensor is a multi-time multilinear object and does not possess the same Chapman–Kolmogorov composition structure as a two-time family. Moreover, the manuscript later states that the theorem has no direct process-tensor analogue. These two descriptions should be reconciled.

10. The opening discussion devotes considerable space to explaining that the construction is not time-reversal symmetric, is not a process-tensor construction, and is not a dynamical speed limit. These qualifications are useful, but they could distract from the main result. The introduction could be streamlined by focusing first on the closest convex-perturbation literature and moving some of the defensive qualifications to the discussion.

11. The dependence on the companion paper "Koopman Spectral Theory for Indivisible Stochastic Processes" could be presented more economically. The central algebraic results of the present manuscript appear largely self-contained, while "Koopman Spectral Theory for Indivisible Stochastic Processes" supplies terminology, regularity conventions, and the interpretation of the integrated deficit. A shorter statement distinguishing the imported results from those proved here would improve readability.

12. Under the column-stochastic convention, the identity

$$\sum_i \Gamma_{ij} = 1 \quad \Longleftrightarrow \quad \Gamma^* \mathbf{1} = \mathbf{1}$$

is correct. Nevertheless, it would help to state explicitly that probability distributions are propagated as column vectors by $\Gamma$, since many stochastic-process texts instead use row-stochastic matrices.

13. The quantity

$$D_{\mathrm{CK}}(\delta t; \Gamma) = \sup_{t-s \le \delta t} D(t, s, r; \Gamma)$$

is a cumulative worst-case deficit over all intervals of duration at most $\delta t$, rather than a deficit at one exact temporal resolution. The definition is

valid, but the phrase 'at resolution $\delta t$" could be misleading. The author should explain this convention or consider describing it as the deficit 'up to temporal scale $\delta t$."

14. Lemma 2.3 and its proof are sound under the stated continuity and boundary assumptions. However, the statement that the deficit map "extends" to $K(\delta t)$ is slightly unnecessary, since the same formula is already meaningful on the equality faces once $\Gamma(u \leftarrow u) = I$ is imposed. It would be simpler to state that the map is continuously defined on the closed set $K(\delta t)$.

15. Definition 2.6 produces a continuous family of column-stochastic matrices, but the manuscript does not show that an arbitrary such convex family is realised as the collection of two-time conditional kernels of one globally defined stochastic process with compatible finite-dimensional distributions. If the paper concerns abstract two-time families only, this should be stated clearly. If a process-level empirical interpretation is intended, the required compatibility assumptions should be discussed.

16. The deficit identity in Theorem 3.1 and its operator-norm consequence are correct. However, the statement that the interaction term is 'second order," whereas the substrate deficit is 'first order," should identify explicitly the perturbation variable with respect to which these orders are understood.

17. The condition

$$M(t, s, r) > D(t, s, r; \Gamma_0)$$

shows that the upper bound in (7) initially increases with $g$, but it does not by itself imply non-monotonicity of the exact deficit. The latter also depends on the operator alignment and cancellation between $A - CE$ and $(C-D)(F-E)$. The discussion surrounding Remark 3.2 should distinguish these two claims more precisely.

18. The distinction between $g_{\text{first}}$ and $g_{\text{term}}$ is one of the more valuable conceptual contributions of the paper, since it accommodates possible non-monotonicity of the perturbed deficit. The definitions use the strict condition $D_{\text{CK}} < \varepsilon$, so the infimum need not itself belong to the relevant set and the deficit may equal $\varepsilon$ at the threshold. The subsequent wording should therefore distinguish consistently between behaviour at and strictly above the thresholds.

19. The bound in Theorem 4.3 is valid but unnecessarily coarse. Before replacing $gM_*$ by $M_*$, the manuscript has

$$D_{\text{CK}}(\delta t; \Gamma_g) \leq (1 - g)\big(D_{\text{CK}}(\delta t; \Gamma_0) + gM_*(\delta t)\big).$$

Solving this quadratic envelope directly gives, for $M_* > 0$, the sharper sufficient estimate

$$g_{\mathrm{term}} \leq 1 - \frac{D_{\mathrm{CK}} + M_* - \sqrt{(D_{\mathrm{CK}} + M_*)^2 - 4M_*\varepsilon}}{2M_*}.$$

The current linear bound could then be retained as a simpler corollary. Moreover, "operationally indistinguishable from a divisible family" should be qualified as indistinguishable according to the chosen CK-deficit statistic, temporal resolution, and sensitivity.

20. Remark 4.6 requires additional justification. For a general irreducible, non-reversible, or non-normal generator, exponential convergence in operator norm may not be governed solely by the spectral gap without further constants or assumptions. In addition, mixing of $\Gamma_{\mathrm{ref}}$ alone does not directly make $M_*$ small, since $M_*$ depends on the distance between the substrate and reference families on both subintervals. The manuscript should state the necessary hypotheses and show explicitly how the cited mixing estimate controls $M_*(\delta t)$.

21. The proof of Proposition 4.9 is correct, but the resulting product inequality

$$g_{\mathrm{first}}(\delta t; \varepsilon)\, \delta t \geq \eta(\varepsilon)\delta t_-$$

does not establish a genuine resolution–coupling scaling law: it follows by multiplying the uniform bound $g_{\mathrm{first}} \geq \eta$ by the assumed positive endpoint $\delta t_-$. Similarly, defining $\gamma_g = g\Lambda$ in Remark 4.11 and multiplying the same inequality by $\Lambda$ does not create an independent dynamical rate bound. The manuscript acknowledges these limitations, but the prominence of the product and rate formulations appears disproportionate to their mathematical content. Remark 4.11 could be shortened or omitted.

22. The numerical calculations in the two-state illustration appear correct, but the example is specified only at one time triple. It verifies the pointwise deficit identity and demonstrates possible non-monotonicity, but it does not construct a continuous two-time family or realise the resolution-level supremum. The pointwise status of the example should be emphasised more clearly, or the example should be extended to a complete family. Accordingly, the claim that the preceding threshold statements apply "verbatim" should be softened.

23. Proposition 5.2 is correct, but causality follows immediately from the retarded kernel and convexity. Moreover, Theorem 5.3 concerns the integrated pointwise deficit and does not appear to require the retarded-operator construction introduced in Section 5.1. The author should clarify what additional role the $L^2([0,T] \times \Omega)$ operator formulation plays in the present paper, or consider shortening this material.

24. Theorem 5.3 is valid, although the description of quadratic suppression should note that $J$ is defined using the squared pointwise deficit, so the rate follows naturally from the $O(1-g)$ pointwise estimate. The bound could also be sharpened by applying Minkowski's inequality before squaring.

25. Proposition 6.1 appears mathematically sound. However, when $g(t,s,x)$ is not continuous, the resulting collection generally falls outside Definition 2.1, which includes operator-norm continuity as part of a two-time family. The terminology should distinguish the unrestricted pointwise construction from a family satisfying the paper's full regularity assumptions.

26. The construction described as a nonlinear-coupling extension is not nonlinear in the transition kernels; it is a monotone reparameterisation $g \mapsto f(g)$ of the same affine family. The results concerning the generalized inverse are correct but add limited substantive content. The material should be described as a reparameterisation result rather than as a nonlinear-coupling extension.

27. Remark 7.1 should replace the statement that exact restoration occurs "only at $g = 1$" with the weaker statement that it is guaranteed at $g = 1$. The results establish $\Delta(\Gamma_1) = 0$, but do not exclude the possibility that the mixed family becomes divisible at an interior coupling through cancellation.

28. The manuscript correctly explains why Proposition 4.9 is not a dynamical speed limit. Given that the similarity is only formal and arises by multiplying a uniform coupling bound by the positive interval endpoint $\delta t_-$, the extended comparison with speed-limit theory appears disproportionate. This discussion could be reduced to a brief qualification.

29. The observer-interaction interpretation should be weakened. The mathematical model modifies the transition family by definition, but the manuscript has not derived whether a physical observer interaction modifies the underlying dynamics, the observation protocol, or both. Similarly, the deficit subspaces of the substrate are independent of $g$, whereas those of the perturbed family need not be. The discussion should distinguish these claims explicitly.